

中國建築工程(香港)有限公司

質 量 管 理 手 冊

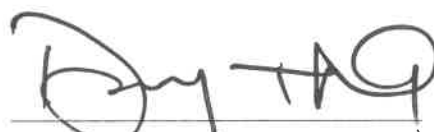
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序 言

為了不斷提高公司工程承包的質量水平，同時為滿足越來越多顧客對質量的要求，公司根據國際標準化組織ISO 9001：2015的要求編制其《質量管理手冊》，並輔以一系列標準工作程序，以切實保證工作質量，滿足客戶要求。

本手冊為中國建築工程(香港)有限公司及其管理之工程公司內部執行文件，版權所有，未經同意不得翻印。手冊之發放由公司綜合管理總監全權負責。

本手冊自簽字之日起正式生效，公司所有員工必須遵照執行。



孔祥兆
董事長兼總經理
二〇二三年四月十二日

《質量管理手冊》修改記錄表

修改日期:2023.04

項目	修改內容簡要	新版
1	第三章增加 QMQE 雙軌制的定義	第四版 B
2	5.3.2 (e) 質量科技部增加對質量管理人員的管理職責	

批准:

董事长兼總經理

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第一章 公司業務與質量管理體系適用及認證範圍

1.1 公司簡介

- 1.1.1 中國建築工程(香港)有限公司(以下簡稱“中建香港或公司”)承接中國建築工程總公司(香港地區)於二〇〇一年轉移之業務，成為香港特區政府建築條例執行處認可房屋工程承建商及發展局五個丙級公共工程牌照，可以競投標額不限的各類公共工程的設計及建造項目，包括房屋工程、土木工程、基礎工程、建築製品工程及室內裝飾等。公司被納入第二組打樁類別專業承建商名冊，亦是香港房屋委員會最大NW2承建商之一。
- 1.1.2 中國海外房屋工程有限公司於一九八八年在香港註冊成立，主要從事房屋建築工程的總承包業務，持有發展局認可公共工程承建商建築類別丙組(試用期)、屋宇處註冊一般建築承建商、基礎和地盤平整專門承建商和香港房屋委員會“NW2”組等牌照。
- 1.1.3 中國建築土木工程有限公司於一九八八年在香港註冊成立，主要從事各類型的土木工程承建業務，包括道路和渠務、水務、地盤開拓及海事工程等，持有工務局認可公共工程承建商道路和渠務乙組、地盤平整丙組(試用期)牌照及屋宇處註冊地盤平整專門承建商。
- 1.1.4 中國建築基礎工程有限公司於一九八六年在香港註冊成立，主要從事基礎工程的設計與施工等業務，於一九八七年一月獲香港特區政府建築條例執行處認可為房屋建築承建商，並獲得香港特區政府的打樁及機械化施工兩項專業執照。
- 1.1.5 中國建築機電工程有限公司於一九九六年六月在香港註冊成立，主要從事空調、通風、電氣、供水、排水、消防、音響、公共天線、防盜保安、緊急發電機及低壓掣櫃安裝等工程。當中包含設計、選購設備、系統安裝、調試、維修及售後保養等環節，持有香港特區政府註冊電業承辦商證明書、註冊專門承建商(通用系統工程)證明書、發展局認可空調裝置裝置(第II組)承造商牌照和電氣裝置(第III組)承造商牌照。
- 1.1.6 愛銘建築(國際)有限公司(原名“AMEX International Construction Limited”)，主要從事房屋及土木工程的设计與施工等業務。
- 1.1.7 海悅建築工程有限公司主要從事樓宇建造、維修、拆卸及裝飾工程。

1.2 公司質量管理體系認證工作範圍

- 1.2.1 公司質量管理體系認證工作範圍包括：
 - a. 地基的設計和建造工程(連續牆、矩形樁、大直徑鑽孔樁、大直徑擴底鑽孔樁、工字鋼樁、鋼管樁、預製混凝土樁、預製預應力管樁、迷你樁、手挖沉箱，套入岩石鋼樁，筏式基腳、擴展基腳和樁帽)，地質勘察工程，及樓宇和結構拆卸工程連拆卸設計；
 - b. 樓宇工程的設計和建造；
 - c. 土木工程的设计和建造(地盤平整、水務、海事、道路和渠務、防止山泥傾瀉/斜坡修補及檔土牆工程)；



- d. 設計、生產和供應混凝土預製件包括預製牆板、樓梯、樓面和護欄；
- e. 採暖系統、機械通風和空調系統、低壓和特低壓電力系統(包括緊急發電機、低壓電制屏櫃、太陽能/風力發電、隔離和不斷電供電系統)、熱水供應系統、供水系統、排水系統、消防系統和廚餘機系統的合約管理、設計、物料供應、安裝、測試、調試、操作和維修保養；
- f. 樓宇保養，維修和改善工程。

1.3 公司質量管理體系各有關認證公司

- 中國建築工程(香港)有限公司
- 中國海外房屋工程有限公司
- 中國建築土木工程有限公司
- 中國建築基礎工程有限公司
- 中國建築機電工程有限公司
- 愛銘建築(國際)有限公司
- 海悅建築工程有限公司



第二章 參考資料

本手冊在編寫過程中主要參考了以下資料：

ISO 9000：2015《品質管理系統－基本認識和詞彙》

ISO 9001：2015《品質管理系統－要求》



第三章 術語

手冊內所採用的術語解釋如下：

術語	解釋
公司	指中國建築工程(香港)有限公司。(本手冊內容同時適用於工程公司)
公司領導	指中建香港的領導人，包括公司執行董事、總經理、副總經理、助理總經理等。
公司綜合管理委員會	由公司指定人士組成的跨部門質量、安全及健康、環境管理機構。
公司綜合管理總監	由公司指定負責管理公司質量、安全及健康、環境等管理體系的人士。
公司質量管理工作小組	由公司指定負責推行公司質量管理制度及措施的跨部門工作小組。
工程部門	指直接從事工程承包與管理業務的房屋工程部、土木工程部、基礎工程部和機電工程部等部門。
職能部門	指行政公關部、質量科技部、人力資源部、安全環保部、物資部、信息化管理部等部門。
工程公司	指中國海外房屋工程有限公司、中國建築土木工程有限公司、中國建築基礎工程有限公司、中國建築機電工程有限公司、愛銘建築(國際)有限公司、海悅建築工程有限公司
單位	指工程部門、工程公司和地盤。
單位總工程師	指相關工程質量及科技負責人，統籌單位質量及科技管理工作。
地盤	指公司和工程公司承建工程的工地。
地盤負責人	指地盤經理。
公司員工	指公司和工程公司及地盤簽約聘請的直屬員工。
設計及開發	指公司和工程公司負責之工程設計及開發。
業主	指以協議或合約形式，將其擁有的建築工程項目交予公司或工程公司承建的人士或機構。
合約	指業主與公司或工程公司之間的正式、有約束力的協議。在協議有效期



內，公司或工程公司將按業主指定要求進行工作。

合約要求	指業主與公司或工程公司所簽定之合約細則及按照此合約發出的指示、工程圖則和技術規範。
分判商	指以合約形式向公司或工程公司承包部份或全部工程的人士或商號。
供應商	指以合約形式向公司或工程公司提供貨品、材料或設備的人士或商號。
產品與服務	指公司及工程公司承建的建築工程和有關之服務。
建築工程	包括房屋、土木、基礎、機電及建築附件的建造工程。
房屋工程	指房屋建造工程，包括住宅建造、商用樓宇及酒店建造、工業廠房及倉庫建造、學校及其它公共建築建造等。
土木工程	指土石方工程、隧道及箱涵工程、海事工程、水務工程、道路及橋樑工程、構築物建設及架設工程、市政工程等。
基礎工程	指基礎建造工程，包括打樁工程、地下連續牆工程、沉箱工程、樁帽建造等。
機電工程	採暖系統、機械通風和空調系統，低壓和特低壓電力系統(包括緊急發電機、低壓電制屏櫃、太陽能/風力發電、隔離和不斷電供電系統)、熱水供應系統、供水系統、排水系統、消防系統和廚餘機系統的合約管理、設計、物料供應、安裝、測試、調試、操作和維修保養；
建築製品工程	指有關建築附件之建造工程，如建造混凝土預製件。
投標期間	指由收到投標文件至交標的一段期間。
合約前期階段	指交標至與業主簽定合約的一段期間。
合約階段	指與業主簽定合約至完成合約的一段期間。
質量管理體系	對質量管理方面，指導和監督公司制定有關政策和目標，並實現有關目標的系統。
質量管理體系審核	審核員客觀地取得審核證據並進行評價，以判斷公司的質量管理體系是否符合公司所規定的質量管理體系審核標準，這是一個系統化、文件化的驗證過程，包括將這一過程的結果呈報公司領導。
質量管理體系所需的流程	包括為各種質量管理活動、資源供應、產品實現，量度等而設的流程。
QMQE雙軌制	質量人員的日常工作表現、培訓、聘用、轉調、考核及辭退由質量科技部(質技部)及工程地盤共同管理的機制。

第四章 公司質量管理體系要求

4.1 總體要求

公司按照國際標準化組織 ISO 9001：2015 的要求，建立、實施和持續地改善其質量管理體系，包括其流程及流程之間的相互作用。公司將：

- a. 確定主要流程投入所需以達至預期效果；
- b. 確定主要流程的次序和相互作用；
- c. 確定及實施必要的準則和方法(包括監控、量度及相關成效指標)，以確保所需流程的有效運作和控制；
- d. 確保獲得所需的資源，以支援這些流程的運作和監測；
- e. 委派有關流程各相關人士，明確職責和權限；
- f. 按 6.1 要求，確定處理有關流程的風險與機會；
- g. 評估這些流程的效果，適當地進行調整或變更，以確保這些流程達致預期結果；
- h. 持續改善這些流程及公司質量管理體系。

4.2 質量管理體系文檔管理

- a. 公司質量管理體系文件見 7.5.1；
- b. 在可行的情況下，公司將：
 - 保持流程運作所需之相關文件及資料；
 - 保留相關資料及記錄，確保這些流程按計劃進行。

4.3 了解相關團體之需要及期望

對於會影響或可能影響公司持續提供符合法例法規及業主滿意的產品及服務之能力的相關團體(如市場競爭主、業主、分判商、供應商等)，公司須：

- a. 明確與公司質量管理體系有關之相關團體；
- b. 確定相關團體與公司質量管理體系有關之要求

公司將監察及覆核相關團體及與其有關要求的資料。

第五章 領導職責

5.1 公司領導承諾

5.1.1 概要

- a. 定期進行領導覆審，明確其對質量管理體系的發展、實施和持續改善其有效性的承諾；
- b. 制定公司質量政策，確立質量目標，符合業主、法例及公司業務營運策略；
- c. 確保公司質量管理體系要求融匯公司業務營運流程內；
- d. 推動流程管理及風險管理的應用；
- e. 確保提供公司質量管理體系所需資源；
- f. 確保傳達公司運作有效質量管理體系的重要性，公司所有員工必須遵照執行，並對工作的質量認真負責；
- g. 確保質量管理體系達致預期的成效；
- h. 制定獎勵辦法鼓勵、指導及支持員工對質量管理體系作出貢獻；
- i. 促進持續改善；
- j. 支持各有關單位牽頭領導做好崗位所需質量管理職責。

5.1.2 客戶為主

公司領導承諾以客戶為主：

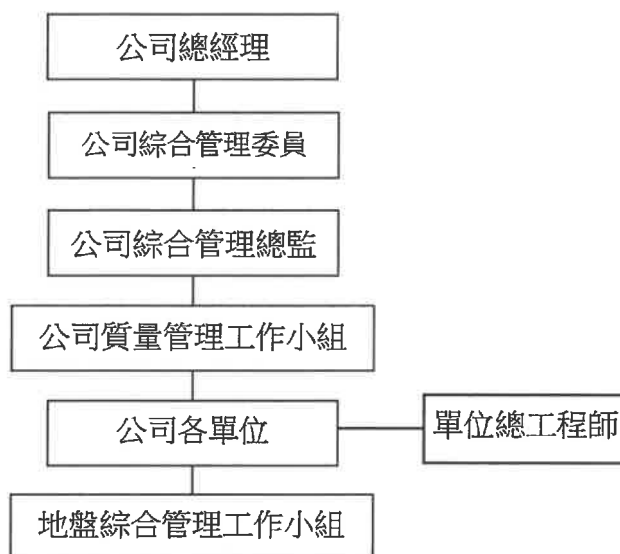
- a. 確定並遵守當地法例法規和業主合約上的要求；
- b. 確定產品與服務上有關風險、機會及提升業主滿意度及提供所需能力；
- c. 以持續提升業主滿意度為主。

5.2 質量政策

- 5.2.1 公司領導制定符合公司業務及營運策略的質量政策，提供框架確立公司的質量目標、指標及有關的義務，並定期評審有關內容，使其不斷切合時宜、有效體現和持續改善質量管理體系及符合有關要求，所有公司員工必須貫徹執行。
- 5.2.2 公司及工程公司質量政策之最新版本，上載公司網址及內網內，供各有關持份者查閱。

5.3 組織架構、職責及權限

5.3.1 組織架構



- 公司領導負責建立、調整及審批公司的組織架構，並將不斷改善，明確所有從事與質量有關工作的人員的責任、權限和相互關係。
- 公司寫字樓各職位的職責及權限由單位負責人會同人力資源部擬定，公司領導批准頒佈。而涉及本手冊的修改則由公司綜合管理總監負責。
- 人力資源部負責各單位最新架構表及公司寫字樓所有職位的職責、權限和相互關係的說明。
- 地盤各職位的職責及權限由地盤經理擬定，編入《項目管理計劃》內。
- 公司《質量管理手冊》及有關附錄文件中明確了資源要求，並保證提供適當的條件和委派經培訓的人員進行管理、執行及驗證工作。如果合約對資源保留有更為嚴格的要求，將在《項目管理計劃》中寫明有關的資源要求，並保證提供適當的條件和委派經培訓的人員。驗證工作包括對設計、施工、安裝和技術服務等過程，已完成工作的檢驗、試驗和監督和內部質量評審。

5.3.2 職責及權限

a. 公司總經理

公司的質量總負責人，負責：

- 釐定及簽署公司質量政策；
- 委任公司綜合管理委員會有關成員及公司綜合管理總監；
- 根據公司質量管理體系落實情況和定期評估結果發出所需指示。

b. 公司綜合管理委員會

公司綜合管理委員會是跨部門質量、安全及健康、環境管理機構，其質量管理主要任務：

- 制定和推行公司的質量政策和措施；

ii. 定期召開會議，研究如何改善公司的質量管理體系和質量水平。

c. 公司綜合管理總監

為公司質量、安全及健康、和環保工作的管理者代表，其負責之質量管理：

- i. 由公司總經理授權，代表公司建立、執行及監管公司質量管理體系，落實國際標準化組織 ISO 9001:2015 版的要求；
- ii. 負責質量管理體系文件的制定、更改和派發；
- iii. 公司質量管理體系的領導覆審會議，向公司領導報告公司質量管理體系的運作、審核及有關目標、指標完成情況和任何改善的需要，並負責貫徹會議精神；
- iv. 協助進行外部審核，並組織和安排內部審核工作；
- v. 確保制定、實施和維持質量管理所需的流程。

d. 公司質量管理工作小組

- i. 協助公司綜合管理委員會推行公司的質量管理措施；
- ii. 建議有關政策和措施；
- iii. 編制及檢討年度質量審核計劃；
- iv. 改善公司的質量管理體系和質量表現。

e. 質量科技部

- i. 貫徹和推行公司的質量政策、目標及指標；
- ii. 建立、推行及發展公司的質量管理體系；
- iii. 負責公司質量管理工作小組的日常工作；
- iv. 負責質量宣傳、推廣、培訓和各項質量評審工作及活動；
- v. 負責監察地盤質量管理體系的運作及表現；
- vi. 收集、統計、分析地盤質量資料及業主滿意度；
- vii 按《地盤質量人員雙軌制管理辦法》管理質量人員的日常工作表現、培訓、聘用、轉調、考核及辭退。

f. 公司各單位

公司各單位包括工程部門、工程公司和地盤等，負責：

- i. 推行公司質量管理體系；
- ii. 執行公司質量管理措施；
- iii. 推動流程及風險管理的應用；
- iv. 按需要發出指示及提供有關工程項目足夠資源及支持。

g. 單位總工程師

- i. 有關工程部門、工程公司質量及科技負責人，統籌有關質量及科技管理工作；



- ii. 負責建立和完善有關工程的質量管理體系，貫徹落實公司質量政策、目標和指標、質量監管措施和相關程序，並確保其有效運行；
 - iii. 配置足夠質量工作人員，包括地盤質量經理、工程師、檢查人員等，明確其質量職責，實施獎勵與懲罰，確保質量監控工作的有效執行；
 - iv. 負責建立工程交付業主前預驗收工作程序、保養期管理制度、以及客戶投訴管理機制，不斷提升業主滿意度；
 - v. 審核項目關鍵監控要求及施工質量問題整改措施；研究工序質量通病制定相關改善措施；
 - vi. 組織執行工程質量檢查及監控工作；
 - vii. 對質量事故進行調查和跟進處理，防止重覆發生；負責組織對質量事故的調查、聆訊、處理，並提出改善措施及督促落實。
- h. 地盤綜合管理工作小組
- 其主要質量管理任務：
- i. 研討改善工程質量；
 - ii. 落實執行施工現場的質量監控工作。

第六章 策劃

6.1 對應風險與機會的相應措施

6.1.1 公司按確立之內外在因素及各有關持份者的要求，確定需考慮的風險與機會，並以下列目標，做好有關策劃工作：

- a. 確保質量管理體系達致預期成效；
- b. 避免或減少不良影響；
- c. 公司表現得到改善。

6.1.2 公司需策劃：

- a. 按照所確立之風險與機會的影響情度制定所需處理措施；
- b. 制定融匯於公司質量管理體系有關流程(見4.1)的方法；
- c. 制定評估有關措施成效的方法。

6.2 質量目標及其完成計劃

6.2.1 公司按不同業主對產品不同的要求，制定質量目標及指標，並保存書面資料。在制定質量目標及指標時，確保：

- a. 與公司質量政策一致
- b. 質量目標和指標具體、明確，可量度；
- c. 考慮各項相關的要求；
- d. 符合產品與服務的質量要求及提升業主滿意度；
- e. 每年檢討一次，並將有關文件分發各有關單位。地盤應將有關文件張貼於地盤的告示板上。

6.2.2 公司為達到質量目標，將確定：

- a. 需完成事項
- b. 所需資源
- c. 所需負責人員
- d. 所需完成時間
- e. 評審結果的成效

6.3 質量管理體系的修改計劃

質量管理體系修改時，公司將制定所需修改計劃並考慮：

- a. 修改目的及其將產生的潛在後果
- b. 修改後對質量管理系統的完整性
- c. 修改後對資源的可用性



- d. 修改後相關人士的責任及權限的分配或重新分配需要

6.4 《項目管理計劃》

為使本《質量管理手冊》所確立的質量管理體系能具體在工程項目上有效地運作，每項工程須根據合約要求制定《項目管理計劃》。內容包括：

- a. 按照工程有關內外因素及有關持份者的要求，地盤需編制《工程質量監控工作計劃》，確定各項潛在質量風險；
- b. 按照工程需要及所確定潛在質量風險，確定所需的工作程序、質量控制措施、施工方案和配備必要的合適人員、試驗及檢驗設備、施工機械及設備等；
- c. 需要時，調整質量控制、檢驗和試驗技術，包括研制新的測試設備；
- d. 當合約要求超出公司現有測量水平時的對策；
- e. 產品落實過程中在適當階段時確定合適的驗證工作；
- f. 寫明合約要求的合格標準；
- g. 要求做好的標記和質量記錄。

《項目管理計劃》的編制，應參照《項目管理計劃編制指導》及《機電安裝管理工作程序》(只適用於機電工程)。第一次編制應重點落實未來 6 個月內的工作計劃。地盤要定期(約每 3 個月)檢討及修改《項目管理計劃》，以確保計劃之有效性。

在地盤完工後，應將《項目管理計劃》的最新版本，以電子檔案記錄，並抄送有關工程單位及質量科技部作為參考及備用。

第七章 資源管理

7.1 資源

7.1.1 概要

公司於每年領導覆審中，檢討內部能力和不足之處，按需要尋求外界支援，確定提供所需資源，以實施、維持和持續改善公司質量管理體系。

7.1.2 人力資源

公司各單位確定和提供足夠人員，以便有效執行公司質量管理體系及各工程施工與監控流程。

7.1.3 基礎設施

各單位需按工程合約的需要，編制工程營運所需設施的清單，確定提供和保養符合產品要求所需的各種運作必需的設施、器材和服務，包括：

- a. 各施工場地、相關的公用設備；
- b. 各施工流程設備如硬件、軟件；
- c. 各類支援服務如運輸、通訊等。

7.1.4 工作環境

各單位均需按照公司、業主的要求，確定、提供及維持對符合產品要求所需的工作環境。

7.1.5 監測和量度設備

7.1.5.1 總則

- a. 地盤應按《施工管理、檢驗和試驗工作程序》及《機電安裝管理工作程序》(只適用於機電工程)中的要求，制定所需執行的監測和量度，以及用作直接證明產品符合要求所需要的監測和量度設備。有關程序已建立符合監測和量度要求的流程和方法。
- b. 公司於《施工管理、檢驗和試驗工作程序》內建立了地盤質量表現監察系統，對表現欠佳地盤按情況發出預警、黃牌及紅牌。

7.1.5.2 量度設備的要求

為確保量度結果的可追溯性及有效性，用作量度的設備須：

- a. 定期或在使用前按量度標準校準和驗證。量度標準應可追溯到國際或國家的標準。若沒有這些標準，用以校準的依據須予以記錄；
- b. 按需要進行調校或再調校；



- c. 予以識別，以便能確定校準狀態；
- d. 避免一些會引致量度結果失效的調校；
- e. 保護其在搬運、保養、儲存當中，免於受損或狀態轉差。

如在施工期間發現設備不符合要求時，須評估和記錄先前量度結果的有效性，並須就設備和任何受影響的產品採取適當的行動。校正和確認的結果須予以記錄存檔。

7.1.6 工程項目施工組織設計及總結

- a. 工程項目施工組織設計按公司《工程項目施工組織設計編制與評審管理辦法》，在開工初期由地盤進行編制並由有關評審小組進行評審，經審批後由地盤組織實施，由有關人員定期進行監督檢查；
- b. 對於項目管理風險高或社會影響範圍廣的大型工程項目投標方案的施工組織設計需進行標前評審工作；
- c. 工程項目施工組織設計如有重大變更，其變更修改由施工組織設計原審批人審核，由地盤技術負責人如地盤代表或建造經理將變更內容及時通知有關單位；
- d. 按公司《科技管理手冊》的要求，大型工程項目完工後由專責部門牽頭組織地盤進行施工技術總結，一般工程項目的施工技術總結工作由地盤組織完成，有關資料呈質量技術存檔及供公司管理人員參考借閱。

7.2 合適人員的安排

公司：

- a. 明確對會影響公司質量管理系統的表現及成效的人員所需具備的能力；
- b. 確保有關人員在教育、培訓、技能和經驗上都有能力勝任；
- c. 按需要，對未具備這些能力的員工，安排接受適當的培訓，或採取其他行動，並評估有關培訓的效果；
- d. 保存有關教育、培訓、技能和工作經驗等記錄；
- e. 人力資源部每年對培訓工作進行一次全面的總結，審議培訓計劃完成情況和培訓的效果，研究需要加強的培訓內容，同時編制新年度的培訓計劃。

7.3 提高和保持質量意識

公司每一員工需對工作質量認真負責，和明白執行以下內容的意義和重要性：

- a. 質量政策；
- b. 有關質量目標及指標；

- c. 員工的參與及貢獻能提升質量管理系統的成效；
- d. 不符合質量管理系統的要求所引致不良後果。

7.4 信息傳遞

各單位按照公司質量管理體系的要求，制定對內及對外所需的溝通，包括：

- a. 公司各單位及各級之間的內部信息傳遞；
- b. 接收、記錄及回應外界的有關信息；
- c. 行政公關部負責按公司領導指示對外發佈及記錄公司有關質量管理的信息。

7.5 質量管理體系文檔要求

7.5.1 公司質量管理體系文件包括：

- a. 公司質量政策；
- b. 公司質量目標和指標；
- c. 本《質量管理手冊》；
- d. 公司《標準工作程序》；
- e. 公司有關質量管理制度、辦法及措施；
- f. 各單位風險評估及編制之質量管理措施；
- g. 工程項目施工組織設計及施工總結；
- h. 《項目管理計劃》；
- i. 質量記錄。

7.5.2 文件編制及更新

質量管理體系所需的文件編制及更新時，須按《文件及資料管理工作程序》的要求，確保：

- a. 附有可識別的特徵，如修改日期、版本等；
- b. 文件在發放前經過審批；
- c. 按需要的格式，以不同形式或媒體有效發放，如CD-ROM、電子郵件，上載內網等。

7.5.3 文件及記錄控制

按《文件及資料管理工作程序》要求，確保：

- a. 在各使用場所備有相應文件的合適版本；
- b. 相應文件妥善存放及保護，按需要確保其保密性、完整性，防止誤用；
- c. 外來文件都加上識別，及做好分發記錄；



- d. 文件保持清晰可讀、易於識別；
- e. 必要時文件將經過評審和更新，並重新審批；
- f. 及時在所有發放和使用場所撤走作廢及過時文件，或採取其它途徑有效防止誤用的情況發生。任何需要保留的作廢及過時文件蓋上識別印章如“作廢”及“保留”等印章。

所有與質量管理有關的記錄須按《文件及資料管理工作程序》建立有關的檔案清單及存放。記錄應清晰可讀、易於識別、所建立的檔案應便於檢索。記錄的識別、存放、保護、取閱、保留期限和棄置所需的控制按《文件及資料管理工作程序》執行。

第八章 工程項目管理

8.1 工程項目施工策劃與控制

各工程項目由投標至完成等所需的流程，需做好策劃、執行及控制，並按第六章所確定之風險與機會執行有關行動，包括：

- a. 確定完成工程項目所要求；
- b. 建立各所需流程的關鍵要素和產品及服務的驗收要求；
- c. 確定所需的資源，以滿足符合產品及服務的驗收要求；
- d. 按所確定的關鍵要素，監控各流程；
- e. 按需要，確定、維持和保留所需記錄，以明確：
 - (1) 有關流程按照計劃進行
 - (2) 產品及服務符合其要求

工程項目施工時所有變更需有計劃控制，對意外而作出的更改按需要進行覆核，以評估所引致的後果，並按需要制定相關措施以減少所引致的不良影響。

公司確保採購、分判及其他外判流程受到監控(見8.4)。

8.2 工程項目施工要求

8.2.1 與業主溝通

各有關單位應按照《文件管理工作程序》、《投標管理工作程序》、《合約管理工作程序》、《物資採購管理工作程序》、《施工管理、檢驗和試驗工作程序》、《設計管理工作程序》及《機電安裝管理工作程序》(只適用於機電工程)等的要求，對：

- a. 承包工程資料；
- b. 工程招標、詢價、合約及有關修改；
- c. 業主的意見，包括業主投訴等；
- d. 業主資產(與工程有關)的管理及控制；
- e. 按需要，為應急行動建立相關要求

與業主進行溝通。

8.2.2 確定與產品相關的要求

公司制定了《投標管理工作程序》及《合約管理工作程序》以評審與產品相關的要求，並於向業主遞交標書、確認合約或確認合約的修改前進行，以確保：

- a. 已界定產品要求：
 - (1). 產品有關的法律和規例
 - (2). 公司確定為所需的要求



- b. 公司能夠應付它所提供的產品之索賠。

8.2.3 覆核與產品相關的要求

- a. 地盤《項目管理計劃》內，需覆核：
 - (1). 業主合約規定的要求，包括交付和交付後活動的要求
 - (2). 業主沒有規定，但為特定用途，或為已知的預期用途所必須的要求
 - (3). 公司確定為所需的要求
 - (4). 與產品有關的法律和規例
 - (5). 與過往合約或訂單要求上的差異

確保所有評審差異進行合適處理，跟進結果按需要記錄存檔。如業主的要求沒有提供書面說明，公司應於簽合約前做好有關確認要求，如可行，應保存記錄。

- b. 評審和跟進結果、產品的新要求等，按需要記錄存檔。

- 8.2.4 若產品要求有任何更改，各地盤及有關部門須確保有關文件已作相應的修改，並通知有關人員。

8.3 產品的設計及開發

8.3.1 概要

質量科技部及有關地盤負責按《設計管理工作程序》的要求，進行設計及開發等有關工作。

8.3.2 設計及開發的策劃

公司質量科技部及有關地盤按各工程承包需要，進行有關設計及開發的策劃和控制，並編制《設計計劃書》，確定：

- a. 設計及開發工作的性質、所需時間和複雜性；
- b. 設計及開發的各個階段，包括設計及開發覆審；
- c. 設計及開發的驗證和確認工作；
- d. 設計及開發各流程負責人員的職責和權限；
- e. 設計及開發的內部及外部所需資源；
- f. 設計及開發流程內有關人員之間相互影響而所需要的控制；
- g. 設計及開發流程業主所需的參與；
- h. 產品交付後的後續要求；
- i. 業主及其他持份者預期對設計及開發流程的參與控制程度；
- j. 相關記錄以證明設計及開發已達到預期要求。

質量科技部及有關地盤負責管理和聯繫參與設計及開發的工作小組，並確保已有效的溝通和責任的明確分配。

設計及開發進行期間，策劃中之設計及開發結果按需要作出相應的更新。

8.3.3 設計及開發的輸入

公司明確產品設計及開發所需的要求，包括：

- a. 設計及開發項目需要達到的功能和性能方面的要求；
- b. 來自以往類似設計的適用資料；
- c. 適用的法例及法規；
- d. 需落實的標準及守則；
- e. 因產品的性質而引致失敗的潛在後果。

輸入的要求須足夠、完備、明確、和無相互抵觸。

設計及開發輸入記錄須保存。

8.3.4 設計及開發的管理

公司對設計及開發流程進行控制，確保：

- a. 明確設計及開發需達致的結果；
- b. 檢討及評估有關達到設計及開發預期結果的能力；
- c. 進行驗證工作，確保設計及開發的輸出達致輸入的要求；
- d. 進行確認工作，確保產品達致應用或使用要求；
- e. 對在檢討、驗證及確認過程中遇到的問題採取必要的措施；
- f. 保留所需記錄。

8.3.5 設計及開發的輸出

公司確保設計及開發輸出達致以下：

- a. 輸入的要求；
- b. 對往後完成產品和服務各流程提供足夠資料；
- c. 包括或參照合適的監控及量度要求，及驗收標準；
- d. 明確產品的特性，提供有用說明預期功能及安全使用方法。

設計及開發輸出的記錄須保存。

8.3.6 設計及開發的更改

公司須確定、檢討及控制產品設計及開發期間或後續的更改，確保不會對產品的符合性造成不良影響。

公司須對以下情況保存記錄：

- a. 設計及開發的更改；
- b. 更改的評審結果；
- c. 設計更改授權人員；
- d. 預防不良影響的措施。



8.4 採購、分判及其他外判的管理

8.4.1 概要

公司須確保採購、分判及其他外判流程及其完成品符合公司要求。

公司制定《物資管理工作程序》及《合約管理工作程序》，對供應商及分判商進行評估及選用，以確保供應商供應的物料及分判商的完成品符合規定的要求，並對以下情況進行監控：

- a. 當供應商的物料及分判商的完成品配合並應用於公司自身的產品上；
- b. 當由供應商及分判商代表公司直接交付產品至業主；
- c. 當公司決定將部份或其中流程由供應商及分判商負責(若有)。

在評估和選用供應商及分判商時，須以他們提供符合公司要求的物料或完成品的能力為依據，評估結果和任何必要的行動須予記錄存檔。

8.4.2 管理方式及範圍

採購物料及分判工程應按《物資採購工作程序》、《合約管理工作程序》、《施工管理、檢驗和試驗工作程序》及《機電安裝管理工作程序》(只適用於機電工程)中的要求進行檢驗、試驗、標識和發放等工作，包括在供應商和分判商的場地執行驗證工作，以確保物料和分判工程符合規定要求。

公司：

- a. 確保供應商和分判商的工作流程在公司質量管理體系控制下運作；
- b. 明確對供應商和分判商，及預期輸出結果有相同監控；
- c. 考慮：
 - (1) 對公司符合業主及法例要求的能力上，供應商和分判商的工作流程所帶來的潛在影響
 - (2) 供應商和分判商自我的監控成效
- d. 明確驗證或有關措施，確保供應物料和分判完成品達到要求。

8.4.3 採購及分判資料

採購及分判資料要清楚地寫明對物料及分判工程的要求，包括審批：

- a. 物料及完成品；
- b. 服務方法、流程和設備及物料與完成品的發放；
- c. 合適人員的資格；
- d. 與供應商和分判商的溝通；
- e. 對供應商和分判商表現的監控；
- f. 於供應商廠房和分判商工場由公司或業主執行的驗證或確認工作。

8.5 施工和服務提供

8.5.1 地盤《項目管理計劃》內應制定必要的控制方法，使在受控的條件下，策劃和進行施工和服務工作。控制包括：

- a. 備有施工所需的資料
 - (1) 產品與服務的施工特性
 - (2) 預期的結果
- b. 備有及使用合適監測及量度設備；
- c. 於適當情況進行施工監測和量度，驗證流程或輸出的控制及產品的驗收符合要求；
- d. 使用合適的設施及於適當的環境下施工；
- e. 委任合資格人士；
- f. 當產品未能透過後續監控及量度進行驗證時，可透過流程的確認及定期檢查，確保產品達致預期成果；
- g. 落實相關措施以避免人為失誤；
- h. 落實發放、運輸及交付後的工作。

8.5.2 標記和追溯

在適當情況下，地盤須按工程施工過程，在《項目管理計劃》中，制定合適的方法及按監測和量度的要求，對各施工階段及產品的狀況，進行可追溯性的標記，並保存有關記錄以確保可追溯性。

8.5.3 屬於業主、分判商或供應商的資產

對屬於業主、分判商或供應商的資產，而交由地盤管理，地盤需加以識別、核對、保護和保管。若有損失、損壞，或不適當使用，應書面通知業主、分判商或供應商並記錄存檔。

8.5.4 產品保存

地盤應於《項目管理計劃》中，制定對各施工階段中，產品的識別、搬運、包裝、儲存和保護，包括半製成品，以確保產品符合要求。

8.5.5 交付後的工作

- a. 公司須符合產品交付後的要求，包括：
 - (1) 法例法規的要求；
 - (2) 產品潛在不希望預到之不良後果；
 - (3) 產品性質、使用及壽命；
 - (4) 業主的要求；
 - (5) 業主的反饋。
- b. 交付後的工作應按《施工管理、檢驗和試驗工作程序》要求：



(1) 若業主合約有規定需要提供工程交付後的服務（缺陷和修補缺陷工作，不屬於交付後服務的範圍），地盤負責人應於項目管理計劃內制定有關交付後服務工作計劃，包括如何執行、監督和檢查交付後服務工作的質量；

(2) 交付後服務工作完成後要進行檢驗或評審，任何不符合合約要求的交付後服務工作，要進行改正和採取預防措施；

(3) 要做好及保存所有有關工作記錄。

8.5.6 變更控制

工程項目施工過程中如有重大變更，其變更修改由施工組織設計原審批人審核，由地盤技術負責人、地盤代表或建造經理將變更內容及時通知有關單位。

公司須保存有關工程項目施工組織設計的評審結果，包括變更的檢討內容、審批人員及處理變更的相關措施。

8.6 產品交付

各工程項目按合約要求，制定各階段施工驗證計劃，進行交付前的各階段驗證工作。除獲得業主許可，各驗證工作應在產品交付業主前完成。

公司須保存有關產品交付的記錄，包括：

- a. 產品符合驗收要求的證明文件
- b. 審批人員的交付記錄

8.7 不符合產品的控制

公司為防止使用或安裝不合格產品，或在工程中留有不合格工作，在《施工管理、檢驗和試驗工作程序》及《機電安裝管理工作程序》（只適用於機電工程）中制定了採購物資和施工工作出現不合格情況下的控制程序。

8.7.1 一般不符合情況的處理方法：

- a. 進行翻工以達到規定要求；
- b. 退回或報廢；
- c. 按需要通知業主；
- d. 經修補或不經修補被讓步接受。

不符合情況經修補，產品須重新驗證以確保達致要求

8.7.2 公司須保留：

- a. 不符合情況記錄；
- b. 相關糾正措施；
- c. 業主讓步接受的記錄；
- d. 確定不符合情況有關負責人員。

第九章 績效評審

9.1 監測、量度、分析及評估

9.1.1 概要

各有關部門按其工作程序的要求及各地盤《項目管理計劃》內應策劃及制定所需的監測、量度、分析和改善等流程及方法，以確保產品符合要求、貫徹及持續改善公司質量管理體系，並保存所需之記錄。

9.1.2 業主滿意度

公司將收集與業主對產品質量之報告、評分等資料進行統計、分析，量度或評估業主的滿意度，並定期覆核有關獲取和分析評估這些資料的方法。

9.1.3 數據分析及評估

公司各單位負責確定、收集和分析合適的數據，以證明質量管理體系的適用性和有效性，和評估可以持續改善質量管理體系的有效性的機會。從監測和量度活動或其他合適來源產生的數據，亦包括在內。

數據分析須提供與下列各項有關的資訊：

- a. 產品要求的符合情況；
- b. 業主滿意度；
- c. 質量管理體系的適用性和有效性；
- d. 按計劃執行成效；
- e. 處理風險與機會之成效；
- f. 供應商、分判商及其他外判的表現；
- g. 質量管理體系持續改善所需。

9.2 內部審核

9.2.1 質量科技部按《質量管理內部審核工作程序》的要求定期進行內部審核工作，以確定公司質量管理體系：

- a. 已符合制定之施工流程、國際標準化組織ISO 9001：2015的要求、及公司所制定的質量管理體系要求；
- b. 已有效地推行和維持。

9.2.2 質量科技部定期制定審核計劃。有關計劃的編制將考慮審核的流程和被審核單位的狀況和重要性，和以往審核的結果。審核的準則、範圍、頻次和方法，按照《質量管理內部審核工作程序》進行，並由與審核內容無直接責任的人進行審核，以確保審核流程的客觀和公正。

被審核單位負責人須確保立即採取行動，以糾正審核發現的不符合情況及其成因。改正措施須包括驗證所採取的行動和報告驗證結果。

9.3 領導覆審



- 9.3.1 公司領導每年對質量管理體系進行有系統的覆審，以確保質量管理體系有效地運作，並滿足國際標準化組織ISO 9001：2015要求和公司的質量政策。

參加人員應包括公司領導、公司綜合管理總監、各有關部門及工程公司負責人、各單位質量總監。

- 9.3.2 公司綜合管理總監負責編制年度報告。報告中應包括：
- a. 上次領導覆審紀要的執行情況；
 - b. 評估與公司質量管理體系有關的外在或內在因素各項變化；
 - c. 評估與公司質量管理體系表現及成效有關的資料，包括：
 - (1) 業主滿意度及各持份者的反饋
 - (2) 質量目標及指標完成情況
 - (3) 流程績效和產品符合情況
 - (4) 不符合和糾正措施情況
 - (5) 監控及量度結果
 - (6) 審核結果
 - (7) 供應商、分判商及其他外判的成效(若有)
 - d. 公司資源提供情況；
 - e. 對確定之風險與機會跟進情況；
 - f. 可改善建議。

- 9.3.3 領導覆審之結論及建議應包括與下列有關的任何決定性行動：

- a. 持續改善建議；
- b. 質量管理體系修改需求；
- c. 資源需要。

- 9.3.4 公司綜合管理總監負責貫徹領導覆審會議精神，落實會議提出的各項任務，並督促和檢查執行情況。

第十章 持續改善

10.1 概要

公司將確定及尋找各項可供持續改善並符合業主要求及提高業主滿意度的機會及採取必要的措施，包括：

- a. 改善產品及工作流程以達致合約要求，並應對未來的需求；
- b. 修正、預防及減省不良影響；
- c. 提升質量管理系統的效能。

10.2 糾正措施

各單位須制定合適的糾正措施來消除不符合的起因，以免這些不符合情況再出現。各單位可參照《施工管理、檢驗和試驗工作程序》及《機電安裝管理工作程序》(只適用於機電工程)的要求：

- a. 評估不符合的情況(包括業主投訴)；
- b. 確定不符合的起因；
- c. 評估是否需要採取糾正措施，以確保不符合不重複出現；
- d. 確定和實施必需的糾正措施；
- e. 評審採取過的糾正措施及其有效性；
- f. 按需要修正公司有關質量管理體系內容；
- g. 記錄採取過的糾正措施及結果。

10.3 持續改善

公司以貫徹執行制定的質量政策、質量目標，並應用審核結果、數據分析、糾正行動及領導覆審等方法，以持續改善質量管理體系的效能。



第十一章 《質量管理手冊》的修改及分發

11.1 《質量管理手冊》的修改

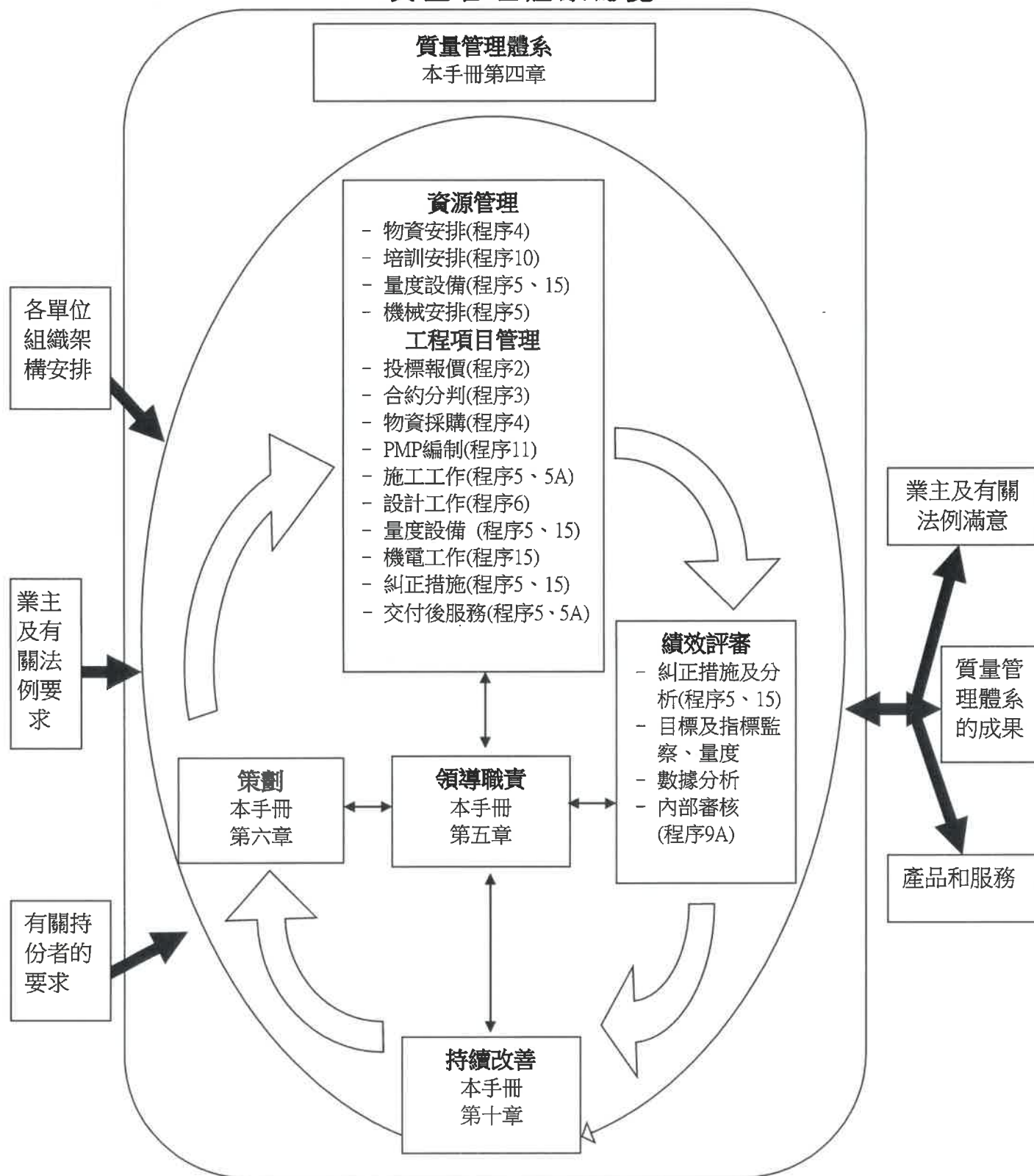
- a. 本手冊由公司綜合管理總監負責修改。沒有正式修改通知，任何修改均屬無效。
- b. 修改情況需記錄在“《質量管理手冊》修改記錄表”上。
- c. 所有修改之內頁的版本編號要按英文字母順序，給予新的編號，日期應為相應的修改日期。
- d. 《質量管理手冊》經一定次數的修改後，應重新印發。重新印發時，應在手冊的面頁註明新版的年份，內頁的版本全部重新改為A版，並註明印發的日期。

11.2 《質量管理手冊》的發放

本手冊以上載公司內網的形式發放。

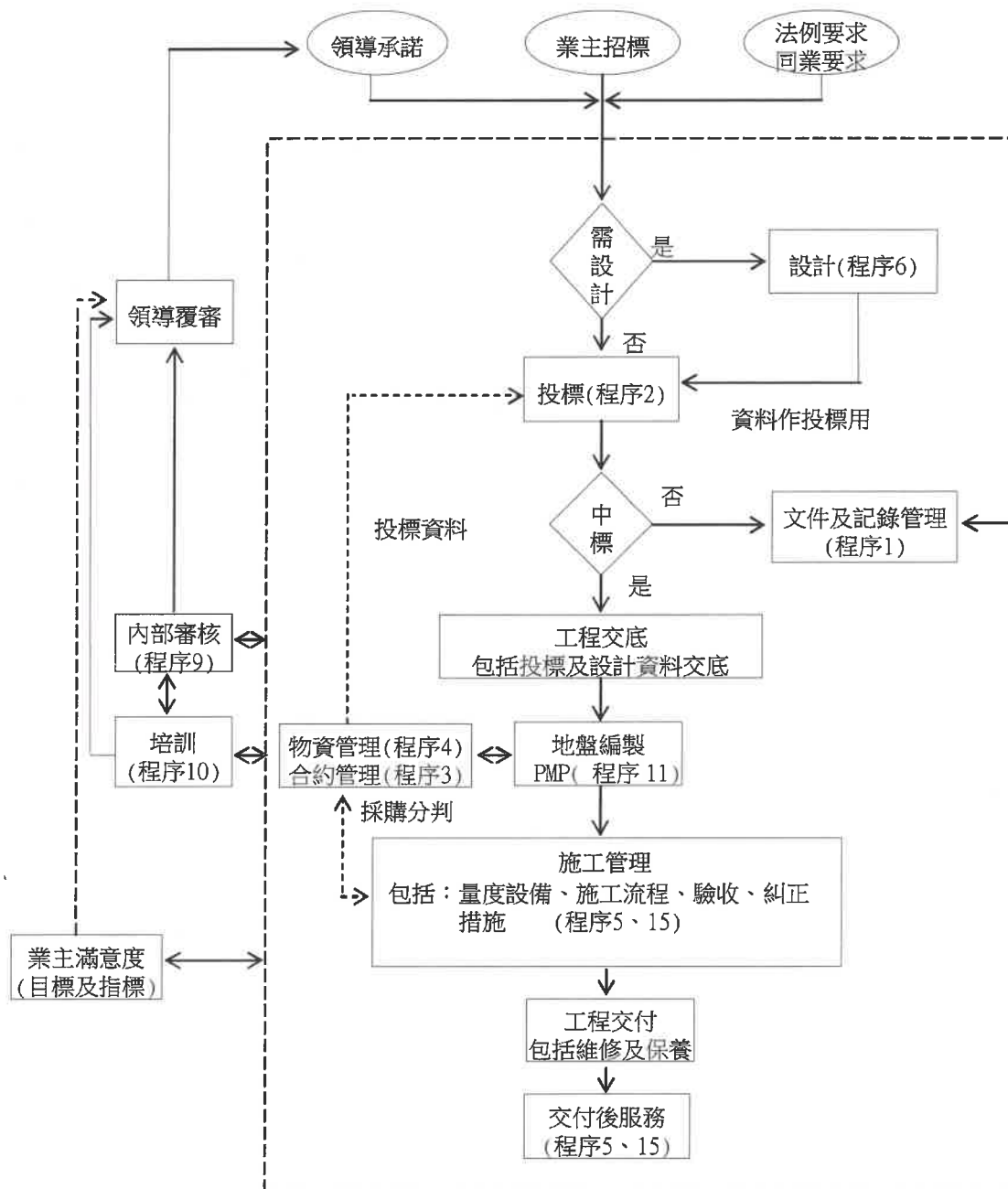
附錄一

質量管理體系總覽



附錄二

質量管理體系內各流程運作概覽



*所有文件及記錄管理按程序1進行

**China State Construction Engineering
(Hong Kong) Limited**

QUALITY MANAGEMENT MANUAL

Revision : 4B
DATE : 2023.05

PREFACE

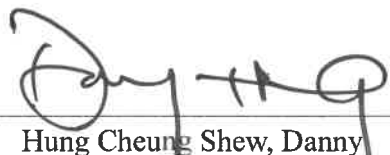
To upkeep the quality of construction works as well as to satisfy the more quality-demanding customers, the Company has compiled its own Quality Management Manual based on the requirement of ISO 9001:2015.

This quality management manual defines the Quality Management System of the Company and is supported by detailed Standard Operation Procedures with aims of enhancing customer satisfaction.

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All staffs shall adhere to this manual effectively from the date of issue.

The English version of this Manual is for translation purpose only. Any inconsistency between English Version and Chinese Version, the Chinese Version shall prevail.



Hung Cheung Shew, Danny
Executive Director & President
10rd May 2023

Amendment Record Sheet

Amended Date : 2023.05

Item	Amendment/Change	New Rev.
1	Chapter 3 adds the definition of QMQE Double-track system	4B
2	5.3.2 (e) The Quality & Technology Department increases the management responsibilities of quality management person	

Approved by:


Executive Director & President

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Chapter 1 Scope of the Company's Business and QMS Application & Certification

1.1 Scope of Company's Business

- 1.1.1 China State Construction Engineering (Hong Kong) Ltd. (CSHK) inherits the business of China State Construction Engineering Corporation (Hong Kong Area) in 2001. CSHK was included in the B.O.O. List of approved building contractors, the Government Lists of specialist contractors for piling works construction. The Company was also granted with the five public work licenses in the Categories C, this is a "confirmed" List II contractor. The Company was thus eligible to submit tenders of unlimited amount in the five major categories of public works. As a result of the above, CSHK has the capacity to undertake a great variety of construction projects including all civil engineering works, foundation works, buildings and internal decoration works. The company is included in the second group of professional contractors in the piling category. It is also one of the largest NW2 contractors of the Hong Kong Housing Authority;
- 1.1.2 COBCL was formed in Hong Kong in 1988. COBCL is engaged mainly in construction of building projects. COBCL was approved contractor of Group C Building Categories of Hong Kong Development Bureau on probationary status. COBCL is also a General Contractor, Foundation and Site Formation specialist in the list of Building Department and NW2 license of the Hong Kong Housing Authority;
- 1.1.3 China State Civil Engineering Ltd. (CSECL) was formed in Hong Kong in 1988. CSECL is engaged in construction of civil engineering projects including roads & drainage works, port works, site formation works and waterworks. CSECL was approved contractor of Group C Civil Engineering Category of Hong Kong Development Bureau on probationary status, and site formation specialist in the list of Building Department ;
- 1.1.4 China State Foundation Engineering Ltd. (CSFE) was formed in Hong Kong in 1986. CSFE is engaged in design and construction of foundation projects, and was registered as general building contractor and foundation specialist contractor of Building Department in January 1987;
- 1.1.5 China State Mechanical and Electrical Engineering Ltd. (CSME) was formed in Hong Kong in June 1996. CSME is engaged in design, supply, installation and maintenance of mechanical ventilation and air-conditioning, electrical, emergency generator, fire services, plumbing and drainage, and extra low voltage systems. CSME is registered electrical contractor and specialist contractor (Ventilation Works) under the registers kept by Government of Hong Kong, and approved specialist contractor under the Air-conditioning Installation (Group II) Category and Electrical Installation (Group III) Category maintained by the Development Bureau;
- 1.1.6 Alchmex International Construction Limited (AICL), originally named as AMEX International Construction Limited, is engaged in design and construction of building and civil engineering projects.
- 1.1.7 Treasure Construction Engineering Limited. is engaged in building construction, maintenance, demolition and decoration works.



1.2 Scope of Works of the Company's Quality Management System

1.2.1 Scope of Works

- a. Design and construction of foundation works (diaphragm walls, barrette pile, large diameter bored pile, large diameter bored pile (with bell out), steel H-pile, steel tubular pile, precast concrete pile, precast pre-stressed tubular pile, mini pile, hand dug caisson work, rock-socketed H-pile, rafts, spread footings and pile cap, provision of Ground investigation work and demolition of buildings and structures with demolition design;
- b. Design and construction of buildings ;
- c. Construction of civil engineering works (port works, waterworks, site formation, roads and drainage) and construction of landslip preventive and remedial works to slopes and retaining walls;
- d. Production and supply of precast concrete products including precast façade, staircase and semi-slab and parapets;
- e. Design, supply, installation contract management and maintenance of heating system, mechanical ventilation and air- conditioning, low voltage and extra low voltage electrical(Including emergency generators, low-voltage electrical screen cabinets, solar/wind power, isolation and uninterruptible power supply systems), hot water supply systems, plumbing, drainage and fire services systems;
- f. Building activities to keep restore and improve the facilities of buildings and surroundings.

1.3 Certified Companies of The Company's Quality Management System

- China State Construction Engineering (Hong Kong) Limited.
- China Overseas Building Construction Limited
- China State Civil Engineering Limited
- China State Foundation Engineering Limited
- China State Foundation Engineering Limited
- Alchmex International Construction Limited
- Treasure Construction Engineering Limited



Chapter 2 References

This Manual is prepared with full reference to the following references:

ISO 9000:2015 Quality Management Systems – Fundamentals and Vocabulary

ISO 9001:2015 Quality Management Systems – Requirements



Chapter 3 Terms and definition

Where the following terms are used within this Quality Management Manual, they shall have the definition so prescribed below:

<u>Term</u>	<u>Definition</u>
The Company	China State Construction Engineering (Hong Kong) Limited
Senior Management of the Company	Includes Executive Director, President, Vice President, Assistant President
Company Integrated Management Committee	Multi-departmental Integrated Management parties, members of which are designated by the senior management of the Company
Company Integrated Management Director	The person designated by the senior management of the Company who is responsible for maintaining the integrated management system of the Company
Company Quality Management Working Group	Multi-departmental Quality Management parties, members of which are designated by the senior management of the Company, responsible to pursue company' s quality management system
Engineering Department	Directly engaged in construction business and management, including Building Construction Department, Civil Engineering Department, Foundation Engineering Department, and Mechanical & Electrical Engineering Department.
Functional Department	Including Administration and Public Relations Department, Quality & Technology Department, Human Resources Department, Safety & Environmental Protection Department, Procurement Department, Information Technology Department etc.
Construction Subsidiaries	China Overseas Building Construction Limited China State Civil Engineering Limited China State Foundation Engineering Limited China State Mechanical and Electrical Engineering Alchmex International construction Limited
Unit	Construction subsidiary, Departments and Sites
Unit Chief Engineer	Responsible for coordinating the quality and technology matters of projects.



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Site	Construction sites of the Company and the Company's subsidiaries
Site Representative	Site Representative means Site Manage
Staff	Person employed directly by the Company or the Company's subsidiaries, or employed by the mean of site contracts
Design & Development	That part of the works for which the Company or the Company's subsidiaries, in accordance with the requirements of the contract, is responsible for producing the design and development
Client	The individual or organization having a Contract with the Company & its subsidiaries for carrying out the construction works (including Mechanical and Electrical Installation works
Contact	Formal and binding agreement between the Client and the Company or the Company's subsidiaries whereby the Company or the Company's subsidiaries is to carry out the works in accordance with the specified requirements of the Client
Contract Requirements	Details of the Contract signed by the Client and the Company or the Company's subsidiaries, in addition to the instructions, drawings and technical specifications Issued under the Contract
Sub Contractors	The individual or organization to whom the company or the company's subsidiaries has contracted part. Or whole of the works
Suppliers	The individual or organization who has a contract with the company or company's subsidiaries for the supply of material, equipment, goods and or services.
Product and Services	Construction works and related services undertaken by the Company
Construction Project	Including Building construction, Civil Engineering, Foundation Engineering, Mechanical & Electrical Installation and Construction product
Building Construction Works	Building construction, comprising residential building, commercial building and hotel, industrial building, warehouse school institutional buildings and other public building, etc.
Civil Engineering Works	Infrastructure works comprising bulk earthmoving,, tunnels and box culverts, maritime works, waterworks, roads, bridges, structures installation and municipal works, etc.
Foundation Works	Foundation works comprising construction of piling, diaphragm walling, caissons and pile caps, etc.



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E&M Works	Heating systems, mechanical ventilation and air conditioning systems, low and extra low voltage power systems (including emergency generators, low voltage electrical panels, solar/wind power, isolated and uninterruptible power supply systems), hot water supply systems, water supply systems, Contract management, design, material supply, installation, testing, commissioning, operation and maintenance of drainage systems, fire protection systems and kitchen waste systems
Construction Product Works	Accessory of Building Construction Works comprising the construction of precast facades, door & aluminum windows, etc.
Tender Stage	The stage from the receipt of tender documents to the submission of the tender
Pre-contract Stage	The period immediately after the submission of tender documents to the receipt of the Client's letter of intent or other instruction to proceed.
Contract Stage	The period from receipt the Client's letter of intent or instruction to proceed till the Issue of final certificate demonstrating the end of the Contract.
Quality Management System (QMS)	Part of the overall management system that includes organizational structure, planning activities, responsibilities, practices, procedures, processes and resources for developing, implementing, achieving, reviewing and maintaining the Quality Policy
QMS Audit	Systematic, documented verification process of objectively obtaining and evaluation audit evidence to determine whether specified Quality Management activities, events, conditions, management systems, or information about these matters conform with audit criteria set by the Company, and communicating the results of this process to the senior management.
QMS Process	Including Processes of Quality Management activities, Resources allocation, Product Realization, Measurement and analysis, etc.
QMQE Double-track system	The daily work performance, training, employment, transfer, assessment and dismissal of quality personnel are jointly managed by the Quality & Technology Department and the engineering site.



Chapter 4 Requirements of Quality Management System for Company

4.1 General Requirement

The Company has established, documented and shall implement and maintain the quality management system, and continually improves its effectiveness in accordance with the requirements of ISO 9001:2015. including the interaction between the processes. The Company shall:

- a. Identify the main process to achieve the desired results;
- b. Determine the sequence and interaction of the processes;
- c. Determine and implement the criteria and methods (including monitoring, measurement and related performance) to ensure that the required processes are effective;
- d. Ensure the availability of resources and information necessary to support the operation and monitoring of the processes;
- e. Delegate all relevant parties in the process, clear responsibilities and authorities;
- f. Refer to Section 6.1 in this manual, identify risks and opportunities to deal with the process;
- g. Evaluate the effectiveness of the processes, appropriately adjusted or changed, in order to ensure that these processes achieve the expected results;
- h. Continuously improve the processes and the company's quality management system.

4.2 Documentations of the Quality Management System

- a. Documentations of the Quality Management System, refer to Section 7.5.1 in this manual;
- b. When feasible, the company will :
 - Maintain the documents and information required for the operation of the process;
 - Maintain relevant information and records to ensure that the processes proceed as planned

4.3 Understand the Needs and Expectations of Interested Parties

For that affect or may affect the company continues to provide interested parties the ability to meet the satisfaction of products and services and owners of laws and regulations (such as the main competition in the market, owners, subcontractors, suppliers, etc.), Company shall

- a. Identify the interested parties related to the company's quality management system;
- b. Determine the requirements of the interested parties related to the company's quality management system.

The company will monitor and review the requirements and information of the interested parties.



Chapter 5 Management Responsibility

5.1 Management Commitment

5.1.1 Summary

- a. Conduct regular leadership of review, to clarify its commitment to the effectiveness of its quality management system for the development; implementation and continuous improvement;
- b. Formulate the company's quality policy and establish quality objectives, in line with the customers, laws and corporate business operations strategies;
- c. Ensure that the company's quality management system requirements are integrated into the company's business operations process;
- d. Promote the application of process management and risk management;
- e. Ensure to provide adequate resources to the company's quality management system;
- f. Ensuring the importance of communicating the company's effective quality management system, all employees of the company must comply with the implementation and accountable of the quality of the work;
- g. Ensure that the quality management system to achieve the expectation;
- h. Develop incentives to encourage, and support employees to contribute to the quality management system;
- i. Promote continuous improvement;
- j. Support the relevant units to process the duties in the quality management.

5.1.2 Customer Focus

- a. The company's leadership is committed to customer-oriented;
- b. Identify and comply with local laws, regulations and requirements of the customer;
- c. Identify risks and opportunities in product or service and to improve customer satisfaction;
- d. Focus on continuously improving customer satisfaction.

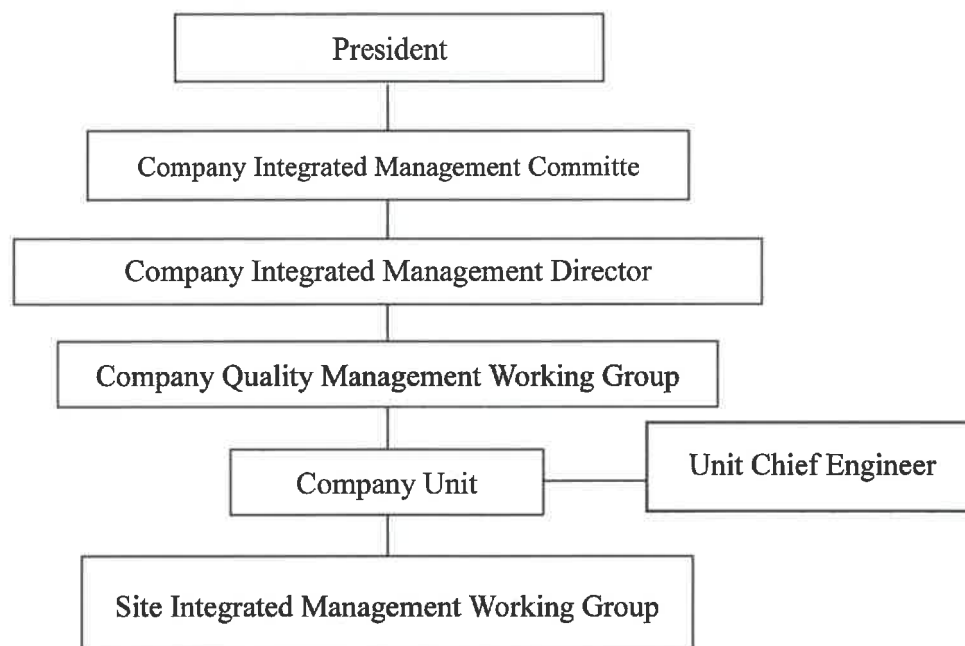
5.2 Quality Policy

- 5.2.1 The Senior Management of the Company has established the Quality Policy and its objectives and targets to comply with requirements and continually improve the effectiveness of the quality management system. The Quality Policy shall be reviewed for continuing suitability.
- 5.2.2 The updated version of Quality Statements of the company and its subsidiaries shall be obtained in our company's intranet.



5.3 Responsibility, Authority and Organization

5.3.1 Organization



The Senior Management of the Company shall maintain a management organization chart, which shall be updated when the management structure changes. The responsibilities, authorities and the interrelation of all personnel who manage perform and verify works affecting quality has been defined. The Senior Management of the Company shall also be responsible for editing, updating and approving the management organization chart.

Job responsibilities and management authorities are prepared jointly by the respective Unit and the Human Resources Department, and reviewed by the Company Integrated Management Director, which are subject to approval by the Senior Management of the Company. Any revision that involves changes to this manual shall be controlled by the Company Integrated Management Director.

The Human Resources Department shall maintain records on the Job responsibilities, responsibility, authority and interrelationship for all staff in the office and updated Construction subsidiaries organization charts.

The Site Representative shall prepare Job responsibilities and management authorities of site staff. Approved positional descriptions and management responsibilities of site staff shall be documented in the respective Project Management Plan.

The Quality Management Manual and its supporting documents have identified the in-house resources requirements and the Company shall provide adequate resources and assign trained personnel for management, performance of work and verification activities. If the resources requirement of a particular Contract is more stringent than the provisions of this manual, the Company shall define in its Project Management Plan the specified resources requirements and provide the necessary resources. Verification activities shall include inspection, test and monitoring of the



design, production, installation and servicing, processes and/or product and internal quality audit.

5.3.2 Responsibility and Authorities

a. President

Overall control of the Company

- i. Establish and sign the Company Quality Policy;
- ii. Appointed members of Company Integrated Management Committee and Company Integrated Management Director;
- iii. Carry out regular meeting, evaluate and improve Quality Management System and to achieve high quality level.

b. Company Integrated Management Committee:

It is an integrated IMS group to co-ordinate among Safety & Environmental and Quality units of the Company;

- i. Prepare and implement Quality Policy and measures;
- ii. Carry out regular meeting, assess and improve Quality Management System and to achieve high quality level.

c. Company Integrated Management Director:

Designated by the President of the Company and is responsible for the establishment, implementation and maintenance of the quality management system of the Company in accordance with the requirements of ISO 9001:2015 standards:

- i. Prepare, amend and issue of the Quality Management Manual and Operation Procedures;
- ii. Prepare, amend and issue of the Quality Management Manual and Operation Procedures;
- iii. Carry out annual management review meeting, report to Senior Management on the performance of the quality management system including audits result, quality objectives, targets and any needs for improvement, ensure all recommendation of management review is implemented;
- iv. Organize and manage both external and internal audit. Ensure that processes needed for the quality management system are established, implemented and maintained.

d. Company Quality Management Working Group

- i. Assist Integrated Management Committee to implement Company Quality Management measures;
- ii. Provide suggestion for the establishment of Quality Policy and measures
- iii. Organize and assess Quality Audit Plan;
- iv. Improve Company Quality Management System and Quality Performance.

e. Quality and Technology Department

- i. Ensure and implement Company's Quality Policy, Objective and Targets;



- ii. Establishes, implement and continually improve Company's quality management system;
- iii. Manage daily operation of Company Quality Management Working Group;
- iv. Manage quality promotion activities, trainings and assessment on Company's quality issues;
- v. Monitor the performance and implementation of quality management system on sites;
- vi. Collect and analyze quality-related information from sites and level of clients' satisfaction.
- vii. According to the "Site Quality Personnel Management Measures (Double-track System)", manage the daily work performance, training, employment, transfer, assessment and dismissal of quality personnel.
- f. Company Units
 - i. Company units include Construction Subsidiaries, Departments and Sites;
 - ii. Implement Company Quality Management System;
 - iii. Carry out Company Quality Management measures;
 - iv. Promote the application process and risk based management give directions and provide sufficient resources support as needed to the project.
- g. Unit Chief Engineer
 - i. Responsible for coordinating the quality and technology matters of projects;
 - ii. Establish and improve the quality management system of individual unit or subsidiary in accordance with the Company's quality policy, objective and targets, and quality monitoring procedure of the Company;
 - iii. Ensure quality monitoring works effectively executed by deploying adequate quality management staff, such as Site Quality Manager, Engineer or Supervisor;
 - iv. Establish Handover Pre-inspection Procedure, Defects Liability Period Management System (DMS) and Customer Complaint Management System;
 - v. Verify critical quality control arrangement of construction works and rectification measures of quality problems carried out to specified requirements; Investigate and establish improvement measures for recurring quality problems;
 - vi. Organize and implement quality inspection and monitoring works;
 - vii. Analyze issue on quality problem, investigate and follow-up action of quality incident to avoid re-occurrence in future.
- h. Site Integrated Management Working Group
 - Key Quality Management task:
 - i. Review improvement of Construction Quality;
 - ii. Carry out on site Quality monitoring work.



Chapter 6 Planning

6.1 Action to address Risks and Opportunities

6.1.1 According to the established internal and external factors and the requirements of the relevant stakeholders, the company considered the risks and opportunities for the following:

- a. Ensure that the quality management system achieves the desired results;
- b. Avoid or reduce the adverse effects;
- c. Achieve and improve Company performance improved.

6.1.2 Company shall be planned

- a. Develop action to address the established risks and opportunities
- b. Formulate method, integrate and implement actions into the company's quality management system processes (see 4.1)
- c. Establish the measures to evaluate the effectiveness of the process

6.2 Quality Objectives and Planning to achieve them

6.2.1 The Company shall ensure the quality objectives, including those needed to meet requirements for customer and product, are documented established at relevant functions and levels within the company and shall consider:

- a. Consistent with company quality policy;
- b. The quality objectives shall be clear tangible and directive. The quality targets shall be measurable and consistent with the quality policy;
- c. Take into account of all relevant requirements;
- d. Fulfil the quality requirements of products and services and enhance customer satisfaction;
- e. The quality objectives and targets shall review on yearly basis and Issue to all units. All sites shall be required to post the up-dated quality objective and targets on its notice board.

6.2.2 In order to achieve the Quality Objective, the company shall determine what:

- a. Objective to be completed;
- b. Resources needed;
- c. Responsible personnel;
- d. Required completion time;
- e. Evaluate the effectiveness of result.

6.3 Quality Management System Review Plan

When the quality management system is review, the company should consider:

- a. The purpose of the modification and its potential consequences;



- b. Integrity of the Quality management system;
- c. Resources availability;
- d. Allocation and re-allocation of responsibilities and authorities of relevant parties.

6.4 Project Management Plan

To ensure that the quality management system can be implemented effectively on all projects, each project shall have its own Project Management Plan, which is prepared in accordance with the Specified Requirements of the Contract. Project Management Plan shall include, but not be limited to:

- a. According to the stakeholders and interested party needs, Site team shall prepare the “Quality Monitoring and Control list for the construction works to identify the relevant potential quality risk;
- b. According to project needs and the identified potential quality risks, determine the desired operating procedures, quality control measures, the construction program and appropriate personnel with the necessary test and inspection equipment, construction machinery and equipment;
- c. The updating, as necessary, of quality control, inspection and testing techniques, including the development of new instrumentation;
- d. The countermeasure when the specified measurement requirement of the Contract exceeds the capability of the Company;
- e. The identification of suitable verification controls at appropriate stage in the product realization;
- f. Establish acceptance standards of the Contract;
- g. The identification and retention of quality records.

The Project Management Plan shall be established according to the Guide to the Compilation of Project Management Plan, and Management Procedures for Mechanical and Electrical Installation (for E&M works only). The Project Management Plan shall be subject to regular reviews and updating firstly after 6 months on commencement of construction works and thereafter every three months to ensure its sufficiency and adequacy.

The updated Project Management Plan shall be recorded in a specified electronic media and copied to respective Unit and Quality & Technology Department upon the completion of the Project.



Chapter 7 Resources Management

7.1 Resources

7.1.1 General

The Company has determined and provided the resources needed, reviewing on yearly basis, to implement and maintain the quality management system and continually improve its effectiveness, and to enhance customer satisfaction by meeting customer requirements.

7.1.2 Human Resources

The Company shall determine and provide all staff necessary to performing work affecting product quality.

7.1.3 Infrastructures

Each individual Unit shall determine, provide and maintain the infrastructure needed to achieve conformity to production and services requirements. Infrastructure includes:

- a. Buildings, workspace, associated utilities;
- b. Process equipment include both hardware and software;
- c. and supporting services such as transport or communication.

7.1.4 Works Environment

Each individual Unit shall determine and manage, according to the requirement of Senior Management and customer, the work environment needed to achieve conformity to product requirement.

7.1.5 Monitoring and Measuring Devices

7.1.5.1. General

- a. The Site shall determine, according to the Process Control, Inspection and Testing Procedures, and Management Procedures for Mechanical and Electrical Installation(for E&M works only) the monitoring and measurement to be undertaken and the monitoring and measuring devices needed to provide evidence of conformity of product to determined requirements;
- b. In accordance with the Process Control, Inspection and Testing Procedures, the Company has established a quality performance monitoring system to ensure that monitoring and measurement can be carried out and are carried out in a manner that is consistent with the monitoring and measurement requirements. Early Warning, Yellow or Red card are issued to those sites with poor quality performance.

7.1.5.2. Measuring Device Requirements

Where necessary to ensure valid results, measuring equipment shall :

- a. Be calibrated or verified at specified intervals, or prior to use, against measurement standards traceable to international or national



measurement standards; where no such standards exist, the basis used for calibration or verification shall be recorded;

- b. Be adjusted or re-adjusted as necessary;
- c. Be identified to enable the calibration status to be determined;
- d. Be safeguarded from adjustments that would invalidate the measurement result;
- e. Be protected from damage and deterioration during handling, maintenance and storage.

In addition, the Site shall assess and record the validity of the previous measuring results when the equipment is found not to conform to requirements. The Site shall take appropriate action on the equipment and any product affected. Records of the results of calibration and verification shall be maintained.

7.1.6 Construction Project Method Statement and Project Conclusion

- a. The project method statement is based on the company's "Project Construction Organization Design Compilation and Evaluation Management Measures". It is prepared by the site at the beginning of the construction and reviewed by the relevant departments. It will be implemented by the site management team after approval;
- b. For the project consist high construction risk or wide social impact, the project construction method shall be approved before bidding;
- c. If there is a major change in the project construction method, the change and modification shall be informed by the site representative or site manager and reviewed by the original approver of the construction method;
- d. Accordance to the [Technical Management Manual], The relevant department shall lead the site management team of the large scale project to complete the conclusion report of the project and shall be kept in the Technical Quality Department.

7.2 Competence Person

Company:

- a. The Company shall determine the necessary competence of all staff performing work affecting the performance and effectiveness of the quality management system;
- b. Ensure all staff are competent on the basis of appropriate education, training skills and experience;
- c. To provide training or take other actions to satisfy these needs;
- d. To maintain appropriate records of education, training, skills and experience;
- e. To evaluate the works on the completion and effectiveness of training annually, along with identifying training content needed to be reinforced and preparing training plan for the coming year.



7.3 Improve and Maintain Quality Awareness

Every employee of the company needs to be responsible for the quality of work and understand the significance and importance of implementing the following contents

- a. Quality Policy;
- b. Relevant Quality Objective and Target;
- c. Employee participation and contribution can improve the effectiveness of the quality management system;
- d. Adverse consequences due to failure to meet the requirements of the quality management system.

7.4 Communication

In accordance with the requirements of the company's quality management system, All units shall formulate internal and external communication requirements, includes :

- a. Internal communicate between the various levels and functions of the Company;
- b. Receive, document and respond to relevant communication from external interested parties;
- c. Based on senior management's advice, Administration and Public Relations Department shall communicate with external parties and document accordingly with respect to the Company's significant quality aspects.

7.5 Documented information requirements of the Quality Management System

7.5.1 The document of Company's Quality Management System includes:

- a. Quality Policy;
- b. Quality Objective and Targets;
- c. Quality Manual;
- d. Standard Operation Procedures(OP);
- e. Relevant Quality Management System, Method and Procedures;
- f. Quality Risk Assessment and Quality Management Procedure by Company Units;
- g. Project Construction Method and its Conclusion;
- h. Project Management Plan;
- i. Quality Records.

7.5.2 Documentation and update

The documentation and updating of the quality management system shall be in accordance with the requirements of the "Document and Data Control Procedures" to ensure :

- a. Identifiable features, such as date modified, version, etc;
- b. Documents shall be approved before distribution;
- c. The documents can be in any form or type of medium such as CD-ROM, e-mail and intranet etc.



7.5.3 Documents and Records Control

In accordance with the requirements of the “Document and Data Control Procedure” to ensure the documents required by the quality management system under control. The procedure shall ensure :

- a. To ensure that relevant versions of applicable documents are available at points of use;
- b. The corresponding documents are properly stored and protected, and their confidentiality and integrity are ensured as needed to prevent misuse;
- c. To ensure that documents of external origin are identified and their distribution controlled;
- d. To ensure that documents remain legible and readily identifiable;
- e. To review and update as necessary and re-approve documents;
- f. The pertinent Issues of appropriate documents are available at all locations; any invalid and/or obsolete documents are promptly removed from all points of Issue or use, or otherwise assured against unintended use. All superseded or retained for any purpose shall apply suitable identification such as chopped with “Superseded” and “For Record”.

The company shall establish and maintain indexing, filing, and disposition of quality records to provide evidence of conformity to requirements and of the effective operation of the quality management system. Records shall remain legible, ready identification, and easily access. The control of the identification, storage, protection, retrieval, retention time and disposition of records shall be carried out according to the “Document and Data Control Procedure”



Chapter 8 Project Management

8.1 Project Planning and Control

The required processes for each project from bidding to completion shall be planned, executed and controlled, and relevant actions shall be carried out in accordance with the risks and opportunities identified in Chapter VI, including;

- a. Identify the needs for completion of the project;
- b. Establish key elements of each required process and acceptance requirements for products and services;
- c. Identify the resources needed to meet the acceptance requirements for products and services;
- d. Control the process in accordance of the establish key elements;
- e. All records shall be kept to ensure:
 - (1). The process is proceeding as planned;
 - (2). Products and services meet their requirements

The projects shall be reviewed as planned under construction. Any project change without planning shall be reviewed as needed to assess the consequences and to develop necessary measures to reduce the adverse effects.

The company ensures that procurement; subcontracting and other outsourcing processes are monitored (see Chapter 8.4).

8.2 Project Construction Requirements

8.2.1 Customer Communication

The Company shall determine and implement effective arrangements in accordance to Document and Data Control Procedures, Tendering Procedures, Contract Management Procedures, Purchase Control Procedures, Process Control, Inspection and Testing Procedures, Design Control Procedures, and Management Procedures for Mechanical and Electrical Installation(for E&M works only) for communicating with client in relation to:

- a. Project information;
- b. Project tenders, inquiries, contracts or order handling, including amendments;
- c. Customer feedback, including customer complaints;
- d. Management and Control the Resources from Client (related to the project);
- e. Establish the emergency procedures as request;

Communicate with the customer.

8.2.2 Identify the relevant requirements of the product

The company has established the Tendering Procedures and the Contract Management Procedure to review the product-related requirements and to proceed with the



submission of the tender, confirmation contract or confirmation of the contract to the customer to ensure:

- a. Identified the requirements of the product:
 - (1). Products related laws and regulations;
 - (2). The company determines the required requirements.
- b. The company is able to meet the claims for the products it provides.

8.2.3 Review the Requirements of the Product

- a. The following requirements shall be reviewed under the project management plan :
 - (1). Requirements specified by the contract, including requirements for delivery and post-delivery activities;
 - (2). Where no customer's specified requirements, but for a particular purpose or intended use for the known requirements necessary;
 - (3). The company determines the required requirements;
 - (4). (Products related laws and regulations;
 - (5). Differences requirements from past contracts or order requirements.

Ensure that all reviewed discrepancies are handled appropriately and follow-up results are documented as needed. If the owner's request does not provide a written explanation, the company should make the relevant confirmation requirements before signing the contract. If feasible, keep records.

- b. Record shall be kept for review and follow up results, new product requirements, etc..

- 8.2.4 If there are any changes to the product requirements, project sites and relevant departments must ensure that the relevant documents have been modified and notify the relevant personnel accordingly.

8.3 Product Design and Development

8.3.1 General

The Quality & Technology Department and project sites shall handle the design and development works in accordance with the "Design Control Procedures".

8.3.2 Design and development planning

The Quality & Technology Department and project sites of the Company shall plan and control the design and development of product. The Department and sites shall establish Design Plan to determine :

- a. The nature, time and complexity of design and development work
- b. Design and development stages, including design and development review;
- c. Design and development verification and validation;

- d. Design and develop the responsibilities and authorities of the responsible personnel of each process;
- e. Internal and external resources required for design and development;
- f. Control required to interact with each other within the design and development process;
- g. Design and development process customer need to participate;
- h. Subsequent requirements after product delivery;
- i. Customer and other interested party expect to be involved in the design and development process;
- j. Relevant records to demonstrate that the design and development have met the expected requirements.

The Quality & Technology Department and project sites shall manage the interfaces between different groups involved in design and development to ensure effective communication and clear assignment of responsibility.

Planning output shall be updated, as appropriate, as the design and development progresses.

8.3.3 Design and development inputs

The company clarifies the requirements for product design and development, including:

- a. Design and development shall achieve the functional and performance requirements;
- b. Information derived from previous similar designs;
- c. Applicable statutory and regulatory requirements;
- d. Implementation of the standard and code of practices;
- e. Potential consequences of failure due to the nature of the product.

Input requirements must be sufficient, complete, clear, and non-reciprocal. Relevant records shall be kept.

8.3.4 Design and Development Management

The Company shall control the design and development process to ensure:

- a. Clear design and development results;
- b. Review and evaluate the ability to achieve the expected results of design and development;
- c. Perform verification to ensure that the output of the design and development meets the input requirements;
- d. Confirmation work to ensure that the product meets application or usage requirements;
- e. Take necessary measures for problems encountered during the review, verification and validation process;



- f. Relevant records shall be kept.

8.3.5 Design and development Output

The company ensures that the design and development output reaches the following:

- a. Input Requirements;
- b. Provide sufficient information to complete the processes of products and services in the future;
- c. Including or referring to appropriate monitoring and measurement requirements, and acceptance criteria;
- d. Defining the characteristics of the product, providing useful instructions for the intended function and safe use.

Records of design and development output shall be kept.

8.3.6 Design and Development Change

The company shall identify, review and control changes during or after product design and development to ensure that it does not adversely affect product compliance.

The company shall keep records of the following conditions:

- a. Design and development change;
- b. Change the review result;
- c. Authorized personnel for design change;
- d. Measures to prevent adverse effects.

8.4 Purchasing and Sub-contracting Management

8.4.1 General

The Company shall ensure purchased product and sub-letting conform to specified requirements.

The Company has established the "Purchase Control Procedures" and the "Contract Management Procedures" to evaluate and select suppliers and sub-contractors to ensure that the materials supplied by suppliers and subcontractors' finished products meet the specified requirements and monitoring is carried out when:

- a. The supplier's materials and subcontractors' finished products are performed in accordance with the Company's specified requirements ;
- b. The supplier and the subcontractor directly deliver the product to the customer on behalf of the Company;
- c. The Company decides to let supplier and sub-contractor to adopt part or part of the process of the Company specified requirements (if any):

The Company shall evaluate and select suppliers and sub-contractors based on their ability to conform with requirements in accordance with the Company's specified requirements. Records of the evaluations and any actions arising from the evaluations shall be maintained.



8.4.2 Management method and scope

Inspections, testing, identification and release of supplier's materials and sub-contract works conformed to specified requirements shall be carried out according to Purchase Control Procedures, Process Control, Inspection and Testing Procedures, and Management Procedures for Mechanical and Electrical Installation (for E&M works only) to:

- a. Ensure that the work process of suppliers and subcontractors operates under the control of the Company's quality management system;
- b. Ensure controls for processes and expected outputs are put in place on suppliers and sub-contractors ;
- c. Consider:
 - (1). Potential adverse impact on the Company arising from supplier's and sub-contractor's ability to perform processes to meet requirements of laws and customers;
 - (2). Effectiveness of supplier and sub-contractor self-monitoring.
- d. Verify or related measures to ensure that the supply of materials and sub-contracted products meet the requirements.

8.4.3 Purchasing and sub-letting information

Purchasing and sub- letting information shall describe the product and contract to be purchased and sub-let, where appropriate, include:

- a. Materials and products;
- b. Service methods, processes, equipment, and the release of materials and products;
- c. Competence , including any required qualification of persons;
- d. Communication with Supplier and Sub-contractor;
- e. Control and monitoring of performance of the supplier and sub-contractor;
- f. Verification or validation performed by the Company or customer at the supplier's factory and subcontractor's workshop.

8.5 Production and Services Provision

8.5.1 Site Project Management Plan shall plan and carry out production and service provision under controlled conditions. Controlled conditions shall include, as applicable:

- a. The availability of information that describes the characteristics of the product;
 - (1). Construction Property of the product or services;
 - (2). Expected result.
- b. The availability and use of monitoring and measuring devices;
- c. Monitoring and measurement at appropriate stages to verify process or output control and product acceptance meets requirements;



- d. Use suitable facilities and work in an appropriate environment;
- e. Assign Competence Person;
- f. When the product fails to be verified through subsequent monitoring and measurement, the process can be confirmed and regularly checked to ensure that the product achieves the expected results;/Implement relevant measures to avoid human error;/Implement the work after issuance, transportation and delivery.

8.5.2 Identification and Traceability

Where appropriate, the Project Management Plan shall states identification of the product by suitable means throughout product realization, the current product status with respect to monitoring and measurement requirements and the way of control and record the unique identification of the product for traceability.

8.5.3 Assets belonging to the Customer, Sub-contractor or Supplier

The Site shall identify, verify, protect and safeguard customer, sub-contractor or supplier assets provided for use or incorporation into the product. If any property is lost, damaged or otherwise found to be unsuitable for use, this shall be reported to the customer, sub-contractor or supplier and records to be maintained.

8.5.4 Preservation of Product

The Project Management Plan shall states procedures to preserve the conformity of product during production and delivery to the intended destination. This preservation shall include identification, handling, packaging, storage and protection. Preservation shall also apply to the constituent parts of a product.

8.5.5 Post-delivery Work

- a. The company must meet the post-delivery requirements, including:
 - (1). requirements of laws and regulations;
 - (2). product potentially not expected to have adverse consequences;
 - (3). product nature, use and duration;
 - (4). customer's requirements;
 - (5). customer feedback
- b. Post-delivery work shall be in accordance with the requirements of "Process Control, Inspection and Testing Procedures":
 - (1). if the customer's contract stipulates that the post-delivery service (defects and repair defects are not included in the scope of post-delivery services), the site representative shall formulate a post-delivery service work plan in the project management plan, including how to implement, supervise and checking the quality of service work after delivery;
 - (2). inspection or review shall be carried out after delivery of the work. Any post-delivery work that does not meet the contract requirements shall be corrected and relevant preventive measures shall be taken.;
 - (3). all relevant work records shall be maintained.

8.5.6 Control of Changes



If there is a major change in the project construction method, the change and modification shall be informed by the site representative or site manager and reviewed by the original approver of the construction method.

The company shall keep the results of the review of the project construction method, including the content of the review, the approval personnel and the relevant measures for handling the changes.

8.6 Product Delivery

According to the contract requirements, each project shall formulate a construction verification plan for each stage and carry out verification work at each stage before delivery. In addition to obtaining the customer's permission, each verification work should be completed before the product is delivered to the customer.

The company must maintain records of product delivery, including:

- a. Proof of product compliance with acceptance requirements;
- b. Approval record of the person authorizing the delivery.

8.7 Control of Nonconforming Product

The company has established the Process Control, Inspection and Testing Procedures, and Management Procedures for Mechanical and Electrical Installation(for E&M works only) to define the controls and related responsibilities and authorities for dealing with nonconforming product, and to ensure that it is identified and controlled to prevent its unintended use or delivery.

8.7.1 For handling minor nonconformity product:

- a. Reworked to meet the Specified Requirement;
- b. Rejected or scrapped;
- c. Notify customers as needed;
- d. Accepted with or without repair;
- e. When nonconforming product is corrected it shall be subject to re-verification to demonstrate conformity to the requirements.

8.7.2 The Company shall maintain:

- a. Nonconforming record;/Related corrective actions;/Customer accepted concession record;
- b. Determine the Responsible person for deciding the action respect of the non-compliance.



Chapter 9 Performance Review

9.1 Monitoring, Measurement, Analysis and Evaluation

9.1.1 General

Each functional department and site shall plan and implement, according to its working procedures and Project Management Plan respectively, the monitoring, measurements, analysis and improvement processes needed to ensure conformity of the product and quality management system and to continually improve the effectiveness of the quality management system

9.1.2 Customer satisfaction

The company will collect data, analyze measure or evaluate the customer's satisfaction with the customer's report on the quality of the product.

9.1.3 Analysis of Data and Evaluation

Each Unit shall determine, collect and analyze appropriate data to demonstrate the suitability and effectiveness of the quality management system and to evaluate where continual improvement of the effectiveness of the quality management system can be made. This shall include data generated as a result of monitoring and measurement and from other relevant sources.

The analysis of data shall provide information relating to:

- a. Conformity to product requirements;
- b. Customer satisfaction;
- c. Applicability and effectiveness of the quality management system;
- d. Perform results as planned;
- e. Dealing with the risks and opportunities;
- f. Performance of the supplier, sub-contractor and the others;
- g. Continuous improvement of the quality management system required.

9.2 Internal Audit

9.2.1 Quality Manager has delegate the Quality & Technology Department to arrange and implement internal audits according to Quality Management System - Internal Audit Procedures regularly to ensure the quality management system :

- a. Conform to the planned working processes, to the requirements of ISO9001: 2015, and to the quality management system of the Company;
- b. Is effectively implemented and maintained

9.2.2 The Quality & Technology Department shall plan the audit program regularly, taken into consideration the status and importance of the processes and areas to be audited, as well as the results of previous audits. The audit criteria, scope, frequency and methods shall be followed according to Quality Management System - Internal Audit Procedures. Selection of auditors and conduct of audits shall ensure objectivity and impartiality of the audit process. Auditors shall not audit their owns



works.

The responsibility of each auditee shall ensure that actions are taken without undue delay to eliminate detected nonconformity and their causes. Follow-up activities shall include the verification of the actions taken and the reporting of verification results.

9.3 Management Review

- 9.3.1 The Senior Management of the Company shall conduct annually systematic review and evaluation on the quality management system in order to ensure its continuing suitability and effectiveness in satisfying the requirements of ISO 9001:2015 International Standard and the Company's stated Quality Policy.

The Senior Management of the Company in conjunction with the Company Integrated Management Director shall carry out the review and where necessary, the appropriate Unit Representative and Quality Directors shall be invited to attend the review

- 9.3.2 The Company Integrated Management Director shall produce an annual report. The report shall comprise the following:

- a. Implementation of the last leadership review summary
- b. Assess changes in external or internal factors related to the company's quality management system
- c. Evaluate information related to the performance and effectiveness of the company's quality management system, including:
 - (1). Customer satisfaction and feedback from interested parties;
 - (2). Completion of the Quality Objective and Target;
 - (3). Process performance and product compliance;
 - (4). Non-compliance and corrective action;
 - (5). Monitoring and Measuring Results;
 - (6). Audit Result;
 - (7). Performance of the Supplier, Sub-contractor and the Other(if any).
- d. Adequacy of the Company resources
- e. Follow-up on identified risks and opportunities
- f. Improvement Suggestion

- 9.3.3 Conclusion and recommendations reached as a result of the management review shall include the following:

- a. Continuous improvement suggestions;
- b. Quality management system modification requirements;
- c. Resources needs.

- 9.3.4 The Company Integrated Management Director is responsible for ensuring that all recommendations reached as a result of management review is implemented. He shall also conduct review and evaluation on the implementation to ensure its suitability and effectiveness.



Chapter 10 Continual Improvement

10.1 General

The company shall determine and identify opportunities for continuous improvement and compliance with the customer's requirements and to improve customer satisfaction and take the necessary measures, including

- a. Improve products and work process to achieve customer requirements and address future needs;
- b. Correct, prevent and mitigate adverse effects;
- c. Improve the performance of the quality management system.

10.2 Corrective Action

Each Unit shall take action according to the Process Control, Inspection and Testing Procedures, and Management Procedures for Mechanical and Electrical Installation (for E&M works only) to eliminate the cause of nonconformity in order to prevent recurrence. Corrective actions shall be appropriate to the effects of the nonconformity encountered including :

- a. Reviewing nonconformity including customer complaints;
- b. Determining the causes of nonconformity;
- c. Evaluating the need for action to ensure that nonconformity do not recur;
- d. Determining and implementing action needed;
- e. Reviewing corrective action taken and its effectiveness;
- f. Revise the company's quality management system content as needed;
- g. Records of the results of action taken.

10.3 Continual Improvement

The Company shall continually improve the effectiveness of the quality management system through the use of the quality policy, quality objectives, audit results, analysis of data, corrective and preventive actions and management review.



Chapter 11 Issue and Amendment of Quality Management Manual

11.1 Amendment Record

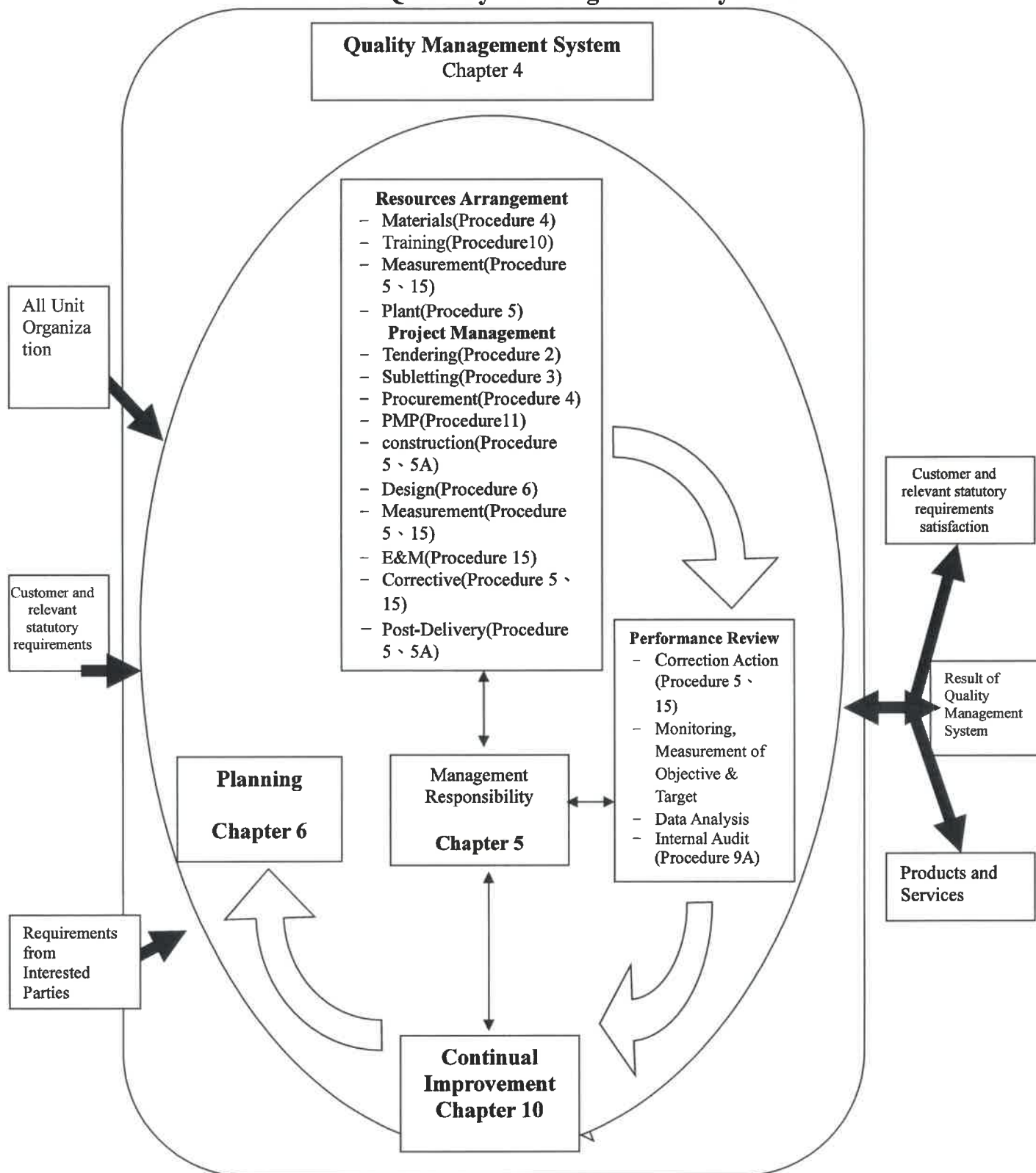
- a. The Company Integrated Management Director is responsible for all updating and/or amendments to this manual. Changes shall not be implemented until the amendments have been formally issued;
- b. Amendment detail shall be described in Quality Management Manual Amendment Record Sheet;
- c. Filled record(s) shall be placed prior to Preface and after cover pages, in sequential order;
- d. After a number of changes, the Quality Management Manual shall be re-Issued. The new version of the manual shall be identified by marking the front page with the year of Issue. The Issue number of the internal pages shall be restarted from A onwards with the new dates of re-Issue.

11.2 Controlled Distribution List

This manual shall be distributed in company intranet.

Appendix I

Model of Quality Management System



Appendix II

Process Flow Chart of Quality Management System

